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First Successful Development Lateral Well in the Early Cenozoic Shammar Clastic Reservoir A Case Study from North Oman Block 5

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Abstract

The newly discovered Shammar Sand in the eastern part of Block 5 was penetrated by four exploration wells, revealing a strong opportunity to expand development of the early Cenozoic reservoir. To guide development with data driven decisions, a horizontal pilot well was drilled to evaluate productivity improvements compared to a vertical pilot hole. Planning and execution were shaped by a series of critical decisions informed by seismic data, well design constraints, and real-time geo-steering challenges.

The evaluation began with a systematic review of gamma ray and mud log data from all nearby wells, including older ones, as each had penetrated the Shammar Sand. This data analysis enabled decision makers to confidently position the first 100 meters within high quality reservoir sand. The reservoir entry point was positioned 15 meters from a well with confirmed Shammar Sand based on log data and directed toward another control point where sand was validated through mud log observations. FMI data further supported trajectory planning and stress regime assessment, ensuring informed decisions for wellbore stability throughout drilling.

A total of 450 meters of sand was drilled, intersecting several sand channels along the well path. The operation was successful, resulting in a prolific production performance three times the vertical well production. By effectively targeting multiple channels with a single horizontal well, the development plan also reduced overall costs by limiting the number of wells required. Few opportunities are also mapped-out to replicate the success of this well. In Addition, the results from horizontal wells will be use to view and update the existing conceptual model.

General seismic evaluation was an integral decision-support tool, involving data conditioning to improve signal quality, followed by a seismic-to-well tie for reflector calibration. Horizon interpretation aimed to map the near-top of the Shammar Sandstone and identify possible Late Cretaceous fault extensions. Key seismic attributes, including RMS amplitude and coherence, were extracted to aid in reservoir characterization. This initial well provided valuable opportunities for the drilling team to validate the well design, optimize mud properties, and determine the most effective BHA configuration and drilling parameters for future wells.

Introduction

Shammar sand one of the youngest hydrocarbon reservoir economically developed in the East of Block 5. Where usually find the Shammar reservoir as a localized thin conglomeratic and porous sand lying directly on top of Late Cretaceous Fiqa Shale and sealed by the Shammar shales. However, toward the western part of Block 5, the Shammar Sand directly overlies the Natih Formation. The Shammar Member is subdivided into four main zones, beginning with a basal conglomerate and sand unit, followed by a clay-rich interval. This is succeeded by a carbonate-cemented sand-rich layer, which is capped by argillaceous limestone at the top (Bazalgette et al., 2021). In some areas, the sand unit is absent and only the conglomerate is present, or the succession begins directly with the clay-rich unit.

The sand unit lies below seismic resolution, and the eastern part of the Block is underlain by Fiqa Shale, which does not offer a preferential or favorable setting for sand to accumulation. The exploration team drilled four wells East of Block 5, confirming the presence of oil, sand and good reservoir quality. The oil encountered was heavier compared to the Western Shammar field in Block 5. To address these limitations and achieve higher productivity, Daleel Petroleum has strategically adopted horizontal well in Shammar. This approach carries greater operational risk especially in maintaining wellbore placement within a narrow reservoir but it offers significant upside in terms of reservoir contact, production potential, and overall development efficiency.

This strategic shift was supported by a high level of integration across multiple disciplines including geology, seismology, drilling, petrophysics, geosteering, and production engineering. The success of the first well resulted from detailed planning, real-time collaboration, and continuous optimization throughout the execution phase. This paper will focus on the technical strategy, execution, and risk management of the first horizontal well highlighting how cross-functional teamwork and strategic thinking have enabled Daleel Petroleum to transform the Sahmmer reservoir from a challenging target into a commercially viable opportunity.

Geological Background

The Shammar Sand was deposited in the early Cenozoic and lies unconformably top a folded and eroded sequence that includes Aptian Shuaiba carbonates, Albian Nahr-Umr shales, Albian to Cenomanian Natih carbonates (Bazalgette et al., 2019) and Coniacian to Campanian foreland basin deposits of Fiqa Shale. The exact stratigraphic position varies depending on the location within the area. The formation consists of a thin, sub-seismic succession of clastic sediments (Bazalgette et al., 2019). The primary reservoir is the basal Shammar sandstone characterized by a thickness ranging from 1 to 6 meters among Block5. Shammar member is the basal part of Umm Er Radhuma formation, which belong to Hashramaut group (Figure 1) (Forbes et al, 2010). Shammar was deposited after Cenozoic unconformity infilling localized erosive lows, and sealed by overlying lower Shammar shales (Al Hajri et al, 2017). In the studied area, Shammar deposited in top of Fiqa shale. The first Alpine regional tectonic event in North Oman, dated to the Upper Cretaceous, was associated with two major phases of ophiolite obduction (Bazalgette et al, 2021). This causing foreland basin and exposed of Cretaceous Nahr Umr shales, Upper Shuaiba carbonates, and Natih carbonates toward Lekhwair area. This event followed by an intense episode of erosion in the Lekhwair area. During the Late Cretaceous, deep-marine Fiqa shales accumulated within the foredeep basin (Boote et al. 1990, Warburton et al. 1990). Following the Cenozoic unconformity, Shammar Sandstone was deposited in certain favorable areas, while conglomerate- and clay-rich units accumulated in others. These deposits were overlain by an argillaceous limestone layer, followed by a carbonate unit, and finally capped with argillaceous limestones at the top (Bazalgette et al, 2021). In the study area, the basal Shammar reservoir units are absent in sectors where the Arada Member is present. In contrast, the lower Shammar conglomerates and sandstones appear to have developed preferentially in areas where Fiqa shales underlie the unconformity.

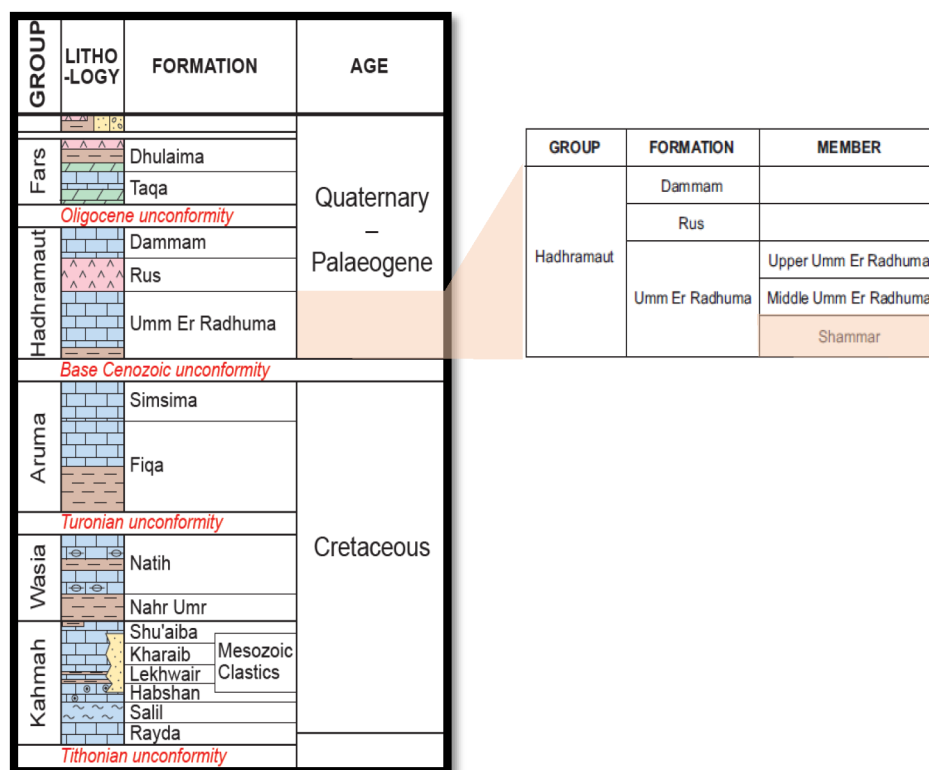


Figure 1—Oman lithostratigraphy, Shammar relative time and related group (Forbes et al, 2010)

Planning, Execution and Operational Challenges

Following the success of four exploration wells in the targeted area, an appraisal lateral well planned to be drill to test the reservoir laterally, to address production challenges associated with heavy oil and low-permeability sand. This well represents the first horizontal development well targeting the Early Cenozoic Shammar clastic reservoir in Block 5. Several challenges encountered during the planning phase, including seismic limitations, reservoir continuity uncertainties, presence of Fiqa shales, drilling mud selection, and understanding reservoir properties and oil saturation. Throughout this process, data-driven analytics became our guiding foundation, anchoring our planning and ensuring that every decision was rooted in strong scientific analysis.

Seismic Imaging

The seismic dataset used in this study comprises a 3D pre-stack depth migrated (PSDM) volume with a temporal sampling interval of 4 milliseconds (ms). Based on the dominant frequency content of approximately 30-35 Hz and an average seismic velocity of around 2,500 m/s in the target zone, the estimated vertical resolution is approximately 15-20 meters, calculated as one-quarter of the dominant wavelength. The Shammar Formation, with a thickness ranging from 1 to 6 meters, falls below this conventional seismic resolution limit, making it difficult to image as a distinct layer. Nevertheless, subtle amplitude anomalies, tuning effects, and well-to-seismic ties enabled indirect detection and mapping of the Shammar sand across parts of the survey area. A time interpretation of a shale horizon (near Shammar reservoir) was carried out, serving as a structural reference for the overlying units. Time-to-depth conversion of this shale surface performed using seismic velocities derived from the 3D PSDM volume. The resulting depth surface calibrated using available well tops to ensure consistency between seismic interpretations and well data. A gross thickness map of the Shammar reservoir subsequently derived from well data. This

thickness map integrated with the calibrated shale horizon depth surface to construct the Shammer reservoir depth map (Figure 2).

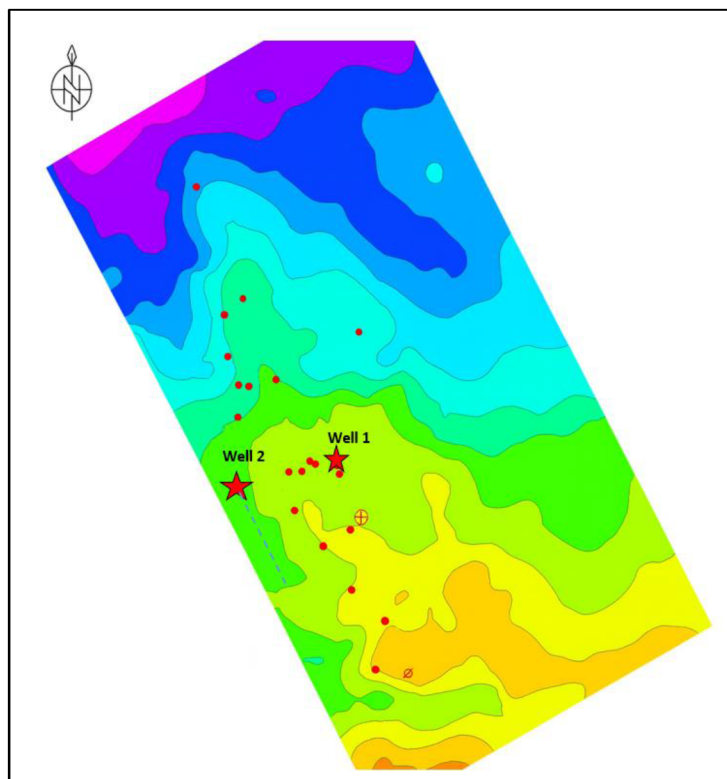


Figure 2—Derived Reservoir Depth Map

The Shammer Sand deposited above a thick shale unit, with no evident structural control on its distribution. Although the area affected by deformed Late Cretaceous faults, there is no clear relationship between faulting and sand deposition. The sand appears to have been deposited independently of the structural framework (due to seismic resolution limitation), likely controlled more by depositional processes than tectonics. Initially the well drilled on structural highs. On the other hand, the current horizontal well intentionally placed in a structurally lower area, where well data indicate good sand thickness and lateral continuity, likely due to stratigraphic rather than structural factors (Figure 3). A deterministic seismic inversion previously attempted in a nearby area where the target sand body is relatively thicker. The objective was to improve the prediction of sand presence and distribution. However, the inversion results failed to clearly delineate the sand, and no consistent impedance response was observed. The thick shale formation underlying the sand posed a major challenge due to its low acoustic impedance contrast with the overlying sands. This, combined with limited seismic resolution and the smoothing effect of the thick shale, masked the sand's impedance response. As a result, the deterministic inversion was unable to reliably map sand distribution (Figure 4).

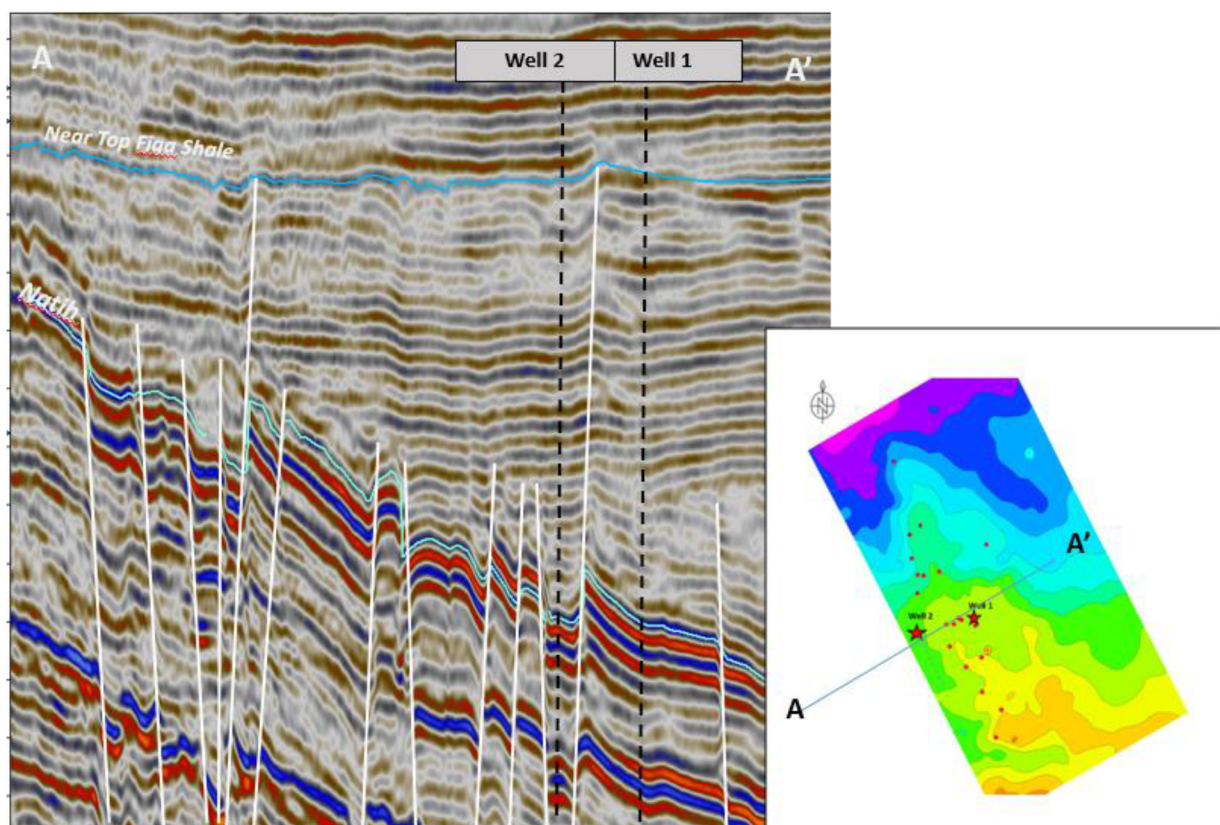


Figure 3—Seismic section across two drilled wells with interpreted faults

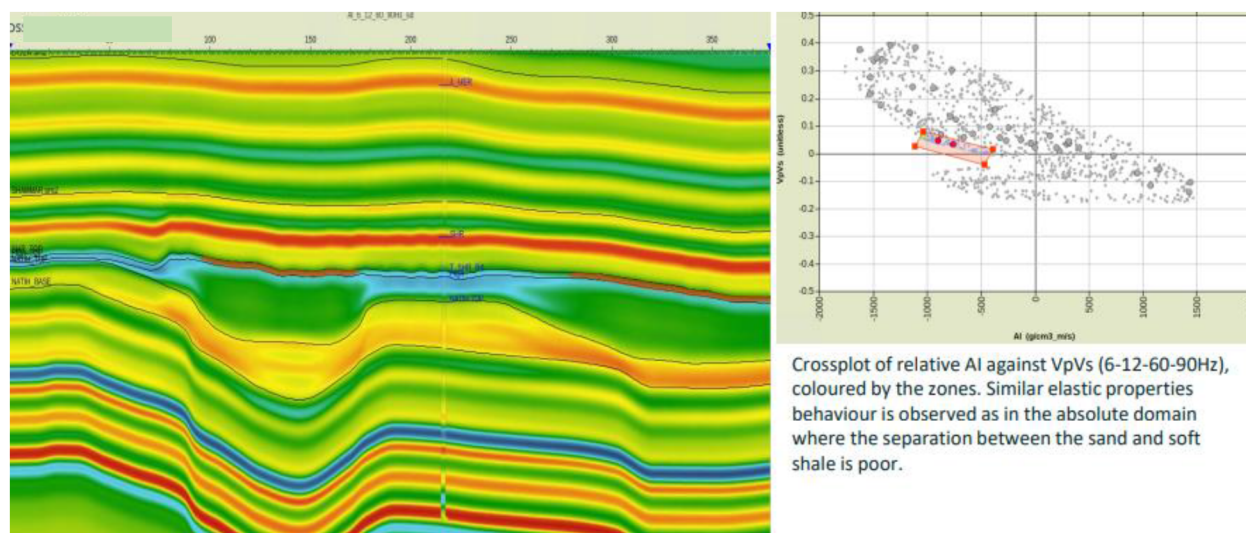


Figure 4—Final modeled Relative Acoustic impedance at certain crossline

Impact of Revised Formation Water Resistivity (R_w) on Saturation Calculation

Petrophysical evaluation plays a crucial role in quantifying hydrocarbon volumes and leading reservoir development strategies. In this study, a critical review of the formation water resistivity (R_w) was undertaken after inconsistencies were observed between the expected R_w from previous knowledge in the field and water salinity from produced wells, Mudlog, Gas shows and core data. Particularly, wells D-1 and D-3 exhibited water cuts of 34% and 80%, respectively with high formation water salinity. Multiple datasets

integrated to re-evaluate the R_w used in the original petrophysical python. Formation Fluid sampling tool: Provided accurate in-situ water salinity and temperature measurements, which were significantly different from pre-drill expectations. Salinity Tests: Laboratory analysis confirmed lower salinity values than previously considered, which directly influencing R_w estimation. Pressure Data and Fluid Gradient: Supported the identification of free water zones and provided independent constraints on saturation interpretation. The revised R_w was calculated using petrophysical software, incorporating actual measured formation temperature and salinity from Formation Fluid samples. Saturation modeling revisited using petrophysical software, applying the updated R_w to Pickett plots (Figure 5) to visualize its effect on S_w distribution across both the sandstone and minor limestone reservoirs. The re-evaluation revealed that the newly derived R_w is approximately 50% lower than the previously assumed value. This adjustment led to: Improved Oil Saturation (S_o) and increasing net pay thickness in both reservoir intervals. Alignment with Core Data oil shows decryption from well D-2, validating the new model. Finally it consistency with Production Data, The high water cut (80%) in well D-3, initially inconsistent with log-derived saturation, is now explained by the corrected S_w profile and fluid gradient data (Figure 6).

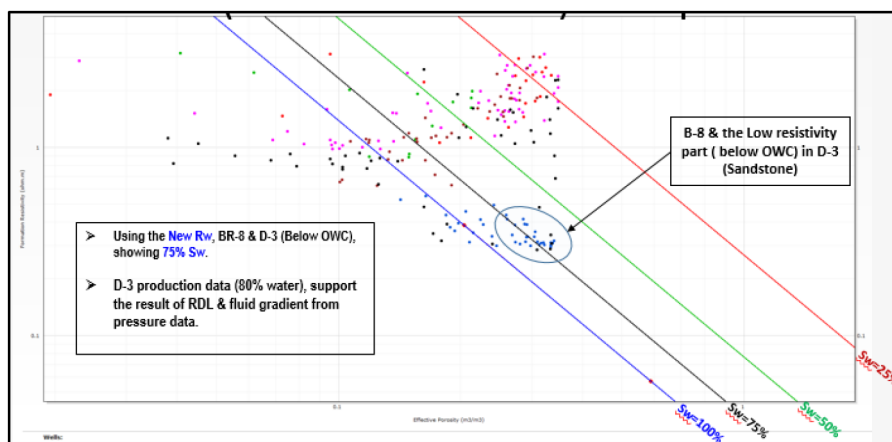


Figure 5—Picket plot (sandstone reservoir) with updated R_w

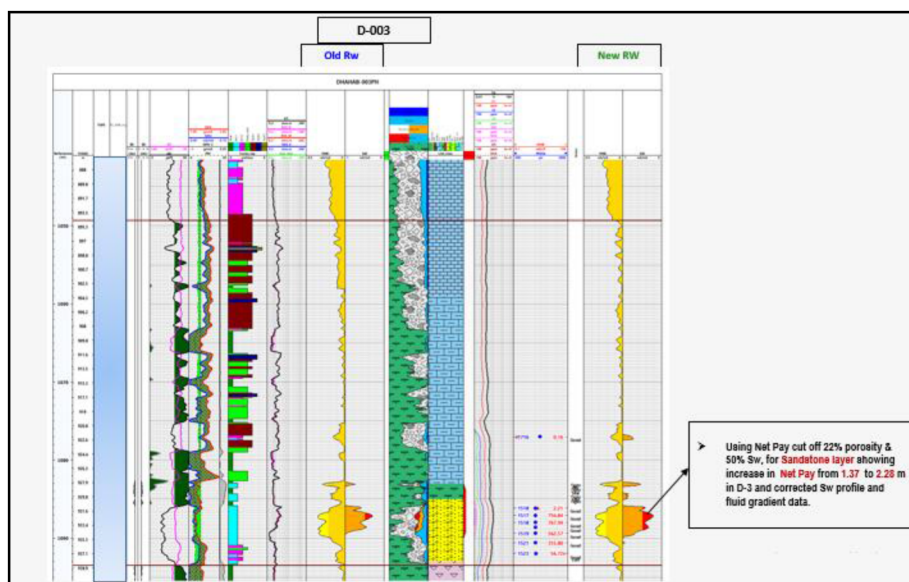


Figure 6—Effect of old vs new R_w in the sand net pay

Mud and Stress Regime

Formation Micro-Imager (FMI) logs and geomechanical study results were critical in planning the first horizontal well across the target sand formation. The FMI provided high-resolution borehole images that identified natural fractures, bedding planes, and breakout zones, which indicated stress orientation and potential wellbore instability risks. These observations integrated with geomechanical modeling, incorporating offset well data to define in-situ stress profiles, pore pressure, and rock strength parameters along the planned well trajectory. Using this combined data, the safe mud weight window (Figure 7) was accurately determined. This ensured that the selected mud weight would be high enough to prevent shear failure and borehole collapse, while avoiding fracturing or losses in weaker zones. The drilling fluid was designed with optimal rheological properties to enhance hole cleaning and cuttings suspension, especially in the horizontal section. A tailored shale inhibitor and fluid loss additives selected, minimizing formation reactivity and filtrate invasion. In addition to careful fluid design, a robust mitigation plan implemented during drilling operations. This included continuous monitoring of torque and drag trends, real-time cuttings return, and flow parameters. After drilling each stand, pickup and slack-off hookloads analyzed for any anomalies that could indicate tight spots or early signs of instability. Immediate corrective actions such as optimizing flow rate, adjusting mud properties, or performing wiper trips were part of the response plan, ensuring borehole stability was maintained throughout the well.

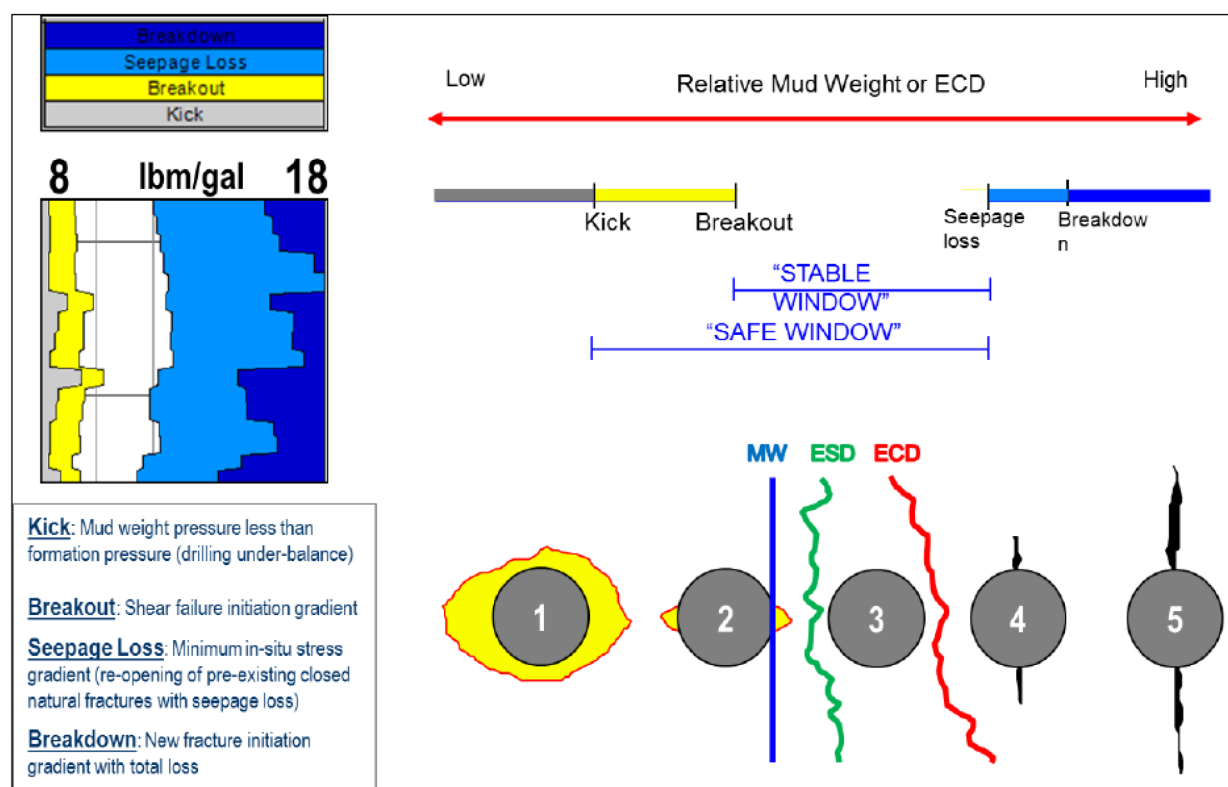


Figure 7—Mud Weight Window

Planning and Operations

Several critical issues needed to be resolved during the planning phase before progressing to execution. One of the primary concerns was that the Shammar Sandstone is sandwiched between two shale Fiqs Shale above and Shammar Shale below which directly influenced the selection of an appropriate mud system to manage shale reactivity and wellbore stability. Additionally, determining the optimal reservoir entry point for the lateral section and steering the well trajectory within the target sand body required careful planning.

Another key consideration was defining the optimum well length within the sand reservoir to maximize production potential while minimizing operational risks.

Shale instability and collapse, particularly while drilling through or in proximity to the Fiqqa Shale formation, represented one of the most critical risks to wellbore integrity. This formation is highly reactive and prone to swelling and sloughing when exposed to drilling fluids, which can rapidly lead to borehole collapse and stuck BHA incidents. To proactively manage this risk, the drilling team conducted real-time monitoring of borehole conditions and implemented responsive mud weight adjustments based on cuttings trends and torque-drag responses. Prior to execution, insights gathered through technical exchanges with other operators who had encountered similar challenges in the region. These discussions helped define a critical time window between drilling and instability onset, allowing the team to optimize tripping and logging operations accordingly.

After drilling four pilot holes, predicting sand distribution and continuity proved to be highly challenging, making it impractical to apply other field's development strategies to be used in this reservoir. Since the primary objective of the well was to test production performance in a horizontal section, it was essential to position the wellbore within a confirmed sand body. To achieve this, a detailed review of all gamma ray and mud log data from the field was conducted to identify the well with the thickest and most continuous sand interval for the reservoir entry (RE) point. Based on this analysis, the RE was positioned 15 meters away from the selected reference well and steered towards another well where the presence of sand had also been confirmed through gamma ray and mud log data. Additionally, FMI data provided insights into the sand deposition direction, allowing the well trajectory to be designed perpendicular to the dip azimuth, optimizing reservoir contact and exposure.

In preparation for execution, the team conducted an in-depth assessment of the optimal drilling strategy. A key discussion centered on whether to deploy a rotary steerable system (RSS) or a motor-based assembly. While RSS offers continuous rotation and consistent LWD data acquisition, previous experience suggested it might struggle to maintain inclination in soft, unconsolidated sands, leading to trajectory control challenges. Conversely, using a motor assembly expected to offer better inclination stability in soft formations. However, this would require frequent sliding to steer the well, which could result in data loss or degradation from the LWD tools, especially azimuthal measurements critical for geosteering. After weighing the trade-offs, the team ultimately decided to proceed with RSS, prioritizing continuous data transmission and smoother hole conditions over the stability benefits of a motor. This decision allowed real-time geosteering decisions to be made with higher confidence, which proved essential in navigating the heterogeneous sand channels. As well as, PeriScope HD was evaluated during the planning phase. This tool offers extended depth of investigation and could have provided valuable insights for steering within the sand bodies. However, due to the lack of prior offset data and the unknown downhole conditions particularly the risk of wellbore instability and potential tool failure the decision made to proceed with a more conventional LWD toolset. The team prioritized wellbore stability and operational reliability over the additional subsurface insight, in order to reduce overall risk to the well.

As this was the first lateral well in the Shammar Sandstone, there was no existing data to define the optimum well length that would match the expected reservoir properties, particularly fluid mobility. Pilot hole results revealed a wide range of mobility readings, varying from low to very high, which introduced significant uncertainty in predicting reservoir performance. Given this variability, the team planned for a lateral section length between 400 to 600 meters, balancing the objective of maximizing reservoir contact with the need to minimize operational risks. The decision also took into account the limited time window for sand exposure before the onset of wellbore instability and collapse risks.

Results and Discussion

The horizontal section was landed in the Shammar sand at the expected depth. A total of 447 meters of lateral section was successfully drilled, encountering multiple sand channels along the trajectory. The structural trend indicated a slight shallowing toward the end of the section. However, the well was terminated earlier than planned due to high surface torque and suspected unconsolidated sand conditions near the toe. Despite not reaching the full planned TD, the well intersected a significant thickness of productive reservoir and confirmed the potential for lateral development in the Shammar sands.

When comparing the production performance of horizontal wells drilled in the Shammar reservoir, to the vertical wells, a clear improvement is observed, despite both wells intersecting zones with similar reservoir characteristics such as porosity, permeability, and net pay thickness. The horizontal well outperformed the vertical well by more than 200% (Figure 8), highlighting the substantial benefits of extended reservoir exposure in Shammar formation. This improvement in productivity is primarily driven by the larger reservoir contact achieved through the horizontal section, which allowed for greater connectivity to high-quality reservoir units and significantly expanded the drainage area. In contrast, the vertical well was limited to a small vertical reservoir contact, resulting in lower flow capacity and restricted pressure support. The production results clearly demonstrate that in similar range of reservoir quality in Shammar formation, horizontal wells development is both technically and economically advantageous. The improvement in productivity confirms that the horizontal approach is the most effective method for unlocking reserves in such formation. The success of the pilot lateral well serves as a proof of concept, encouraging wider application of horizontal drilling across the field and helping to reduce the number of wells needed to achieve development targets.

In parallel, the team developed a notional pre-development strategy to guide future field expansion and define the optimal distribution of horizontal wells across the reservoir. While this conceptual plan was essential in aligning early technical decisions, its details and associated development scenarios will be addressed in a separate, dedicated study.

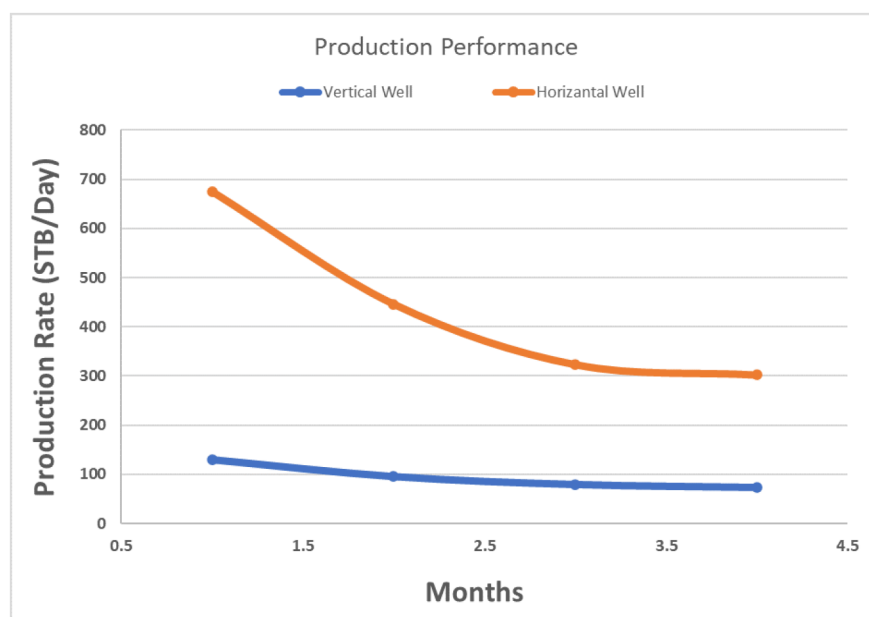


Figure 8—Horizontal well performance vs vertical well performance

Recommendations

From appraisal point of view, it is recommended to reprocess the PSDM seismic data using Full Waveform Inversion (FWI) to improve shallow formation imaging and velocity model accuracy. Additionally, conducting a shear velocity survey is advised to enable stochastic inversion, enhancing sand distribution prediction and reservoir characterization in thin-bed, low-contrast intervals. As the existing mud composition delivered optimal results, reapplying the same formulation is advised. For planning improvement, to utilize resistivity logs while landing to ensure accurate landing within the targeted sand zone. Advanced geosteering technologies should also be utilized to optimize well placement and provide more detailed reservoir mapping for future modeling.

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